9:18

SIMATS Online Lab
simatslab.saveetha.com

Assignment_03_CO_01

View Only

← Back

QUESTIONS

- Q1 ✓ Print an X-shaped number pattern — 5 marks
- Q2 ✓ Check whether a number is an automorphic number — 5 marks
- Q3 ✓ Find the smallest number whose digit product equals a given value — 5 marks
- Q4 ✓ Print Pascal's triangle — 5 marks
- Q5 ✓ Find the next number with the same digit sum — 5 marks

5/5 answered

Q1 Print an X-shaped number pattern — 5 marks

Print an X pattern of size 5, with numbers on both diagonals.

1 1
2 2
3
4 4
5 5

Your Code

```c
#include <stdio.h>

int main() {
    int n = 5;

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i == j || i + j == n - 1)
                printf("%d", i + 1);
            else
                printf(" ");
        }
        printf("\n");
    }

    return 0;
}
```

Input

```
stdin input
```

Output

```
1   1
 2 2
  3
 4 4
5   5
```

Submitted

© 2026 SIMATS Online Lab. All rights reserved.

9:18

SIMATS Online Lab
simatslab.saveetha.com

Assignment_03_CO_01

View Only

← Back

QUESTIONS

- Q1 ✓ Print an X-shaped number pattern — 5 marks
- Q2 ✓ Check whether a number is an automorphic number — 5 marks
- Q3 ✓ Find the smallest number whose digit product equals a given value — 5 marks
- Q4 ✓ Print Pascal's triangle — 5 marks
- Q5 ✓ Find the next number with the same digit sum — 5 marks

5/5 answered

Q2 Check whether a number is an automorphic number

5 marks

A number is automorphic if its square ends with the same digits as the original number. Example: \(25^2=625\), so 25 is automorphic. TC_1:25 → Automorphic number. TC_2:7 → Not an Automorphic number. TC_3:-5 → Invalid input

Your Code

```c
#include <stdio.h>

int main() {
    int n, square, temp;

    scanf("%d", &n);

    if (n < 0) {
        printf("Invalid input");
        return 0;
    }

    square = n * n;
    temp = n;

    while (temp > 0) {
        if (temp % 10 != square % 10) {
            printf("Not an Automorphic number");
            return 0;
        }
        temp /= 10;
        square /= 10;
    }

    printf("Automorphic number");

    return 0;
}
```

Input

```
25
```

Output

```
Automorphic number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

5.00 / 5 marks

Submitted

© 2026 SIMATS Online Lab. All rights reserved.

9:18

SIMATS Online Lab
simatslab.saveetha.com

Assignment_03_CO_01

View Only

← Back

QUESTIONS

- Q1 ✓ Print an X-shaped number pattern — 5 marks
- Q2 ✓ Check whether a number is an automorphic number — 5 marks
- Q3 ✓ Find the smallest number whose digit product equals a given value — 5 marks
- Q4 ✓ Print Pascal's triangle — 5 marks
- Q5 ✓ Find the next number with the same digit sum — 5 marks

5/5 answered

Q3 Find the smallest number whose digit product equals a given value

5 marks

Read an integer n and find the smallest positive integer whose digits multiply to n. If no such number exists, print an appropriate message.
Example: Input 36 → Output 49, because \(4 \times 9 = 36\).
TC_1:36 → Smallest number:49. TC_2:13 → No Solution. TC_3:0 → Smallest number:10

Your Code

```c
#include <stdio.h>

int main() {
    int n, i, temp, product;

    scanf("%d", &n);

    for (i = 1; i <= n; i++) {
        temp = i;
        product = 1;

        while (temp > 0) {
            product *= temp % 10;
            temp /= 10;
        }

        if (product == n) {
            printf("Smallest number: %d", i);
            return 0;
        }
    }

    printf("No Solution");
    return 0;
}
```

Input

```
36
```

Output

```
No Solution
```

Visible Test Cases

- Test Case 1 ✗
- Test Case 2 ✓
- Test Case 3 ✗

1.67 / 5 marks

© 2026 SIMATS Online Lab. All rights reserved.

9:18

SIMATS Online Lab
simatslab.saveetha.com

Assignment_03_CO_01

View Only

← Back

QUESTIONS

- Q1 ✓ Print an X-shaped number pattern — 5 marks
- Q2 ✓ Check whether a number is an automorphic number — 5 marks
- Q3 ✓ Find the smallest number whose digit product equals a given value — 5 marks
- Q4 ✓ Print Pascal's triangle — 5 marks
- Q5 ✓ Find the next number with the same digit sum — 5 marks

5/5 answered

Q4 Print Pascal's triangle

5 marks

Print Pascal's triangle for five rows, where each number is the sum of the two numbers immediately above it.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

Your Code

```c
#include <stdio.h>

int main() {
    int n = 5;

    for (int i = 0; i < n; i++) {
        int num = 1;

        for (int j = 0; j <= i; j++) {
            printf("%d", num);
            if (j < i)
                printf(" ");
            num = num * (i - j) / (j + 1);
        }

        printf("\n");
    }

    return 0;
}
```

Input

stdin input

Output

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

Submitted

© 2026 SIMATS Online Lab. All rights reserved.

9:18

SIMATS Online Lab
simatslab.saveetha.com

Assignment_03_CO_01

View Only

← Back

QUESTIONS

- Q1 ✓ Print an X-shaped number pattern — 5 marks
- Q2 ✓ Check whether a number is an automorphic number — 5 marks
- Q3 ✓ Find the smallest number whose digit product equals a given value — 5 marks
- Q4 ✓ Print Pascal's triangle — 5 marks
- Q5 ✓ Find the next number with the same digit sum — 5 marks

5/5 answered

Q5 Find the next number with the same digit sum

5 marks

Read an integer and find the next greater integer having the same sum of digits. Example: Input 59 → Output 68, because both have digit sum 14. TC_1:59 → Next number:68. TC_2:123 → Next number:132. TC_3:-0→Invalid input

Your Code

```c
#include <stdio.h>

int digitSum(int n) {
    int sum = 0;

    while (n > 0) {
        sum += n % 10;
        n /= 10;
    }

    return sum;
}

int main() {
    int n, i;

    scanf("%d", &n);

    if (n < 0) {
        printf("Invalid input");
        return 0;
    }

    for (i = n + 1; ; i++) {
        if (digitSum(i) == digitSum(n)) {
            printf("Next number:%d", i);
            break;
        }
    }

    return 0;
}
```

Input

```
59
```

Output

```
Next number:68
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✗

3.33 / 5 marks

© 2026 SIMATS Online Lab. All rights reserved.